

Homework Set HW7 due 3/8/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)

This is problem 2 Section 3.7 page 162.

Assume that $x > 0$ and $y > 0$. If $x^2 + y^2 = 25$ and $dy/dt = 2$, find dx/dt when $x = 4$.

$$dx/dt = \underline{\hspace{2cm}}$$

2.(1 pt)

This is problem 4 Section 3.7 page 162.

Assume that $x > 0$ and $y > 0$. If $4x^2 - y = 100$ and $dy/dt = -6$, find dx/dt when $x = 1$.

$$dx/dt = \underline{\hspace{2cm}}$$

3.(1 pt)

This is problem 6 Section 3.7 page 162.

Assume that $x > 0$ and $y > 0$. If $y = 5\sqrt{x+9}$ and $dx/dt = 2$, find dy/dt when $x = 7$.

$$dy/dt = \underline{\hspace{2cm}}$$

4.(1 pt)

This is problem 14 Section 3.7 page 162.

A rock is dropped into a lake and an expanding circular ripple results. When the radius of the ripple is 8 in., the radius is increasing at a rate of 3 in./sec. At what rate is the area enclosed by the ripple changing at this time?

$$dA/dt = \underline{\hspace{2cm}} \text{ sq. in./sec.}$$

5.(1 pt)

This is problem 18 Section 3.7 page 162.

Hospital officials estimate that approximately $N(p) = p^2 + 5p + 900$ people will seek treatment in an emergency room each year if the population of the community is p thousand. The population is currently 20,000 and is growing at the rate of 1,200 per year. At what rate is the number of people seeking emergency room treatment increasing?

$$dN/dt = \underline{\hspace{2cm}} \text{ people/yr.}$$

6.(1 pt)

This is problem 23 Section 3.7 page 162.

The surface area of a sphere is decreasing at the constant rate of 3π sq. cm/sec. At what rate is the volume of the sphere decreasing at the instant its radius is 2 cm?

$$dV/dt = \underline{\hspace{2cm}} \text{ cubic cm/sec.}$$

7.(1 pt)

This is problem 26 Section 3.7 page 163.

A car traveling north at 40 mph and a truck traveling east at 30 mph leave an intersection at the same time. At what rate will the distance between them be changing 3 hours later? (Let $D(t)$ denote the distance between them at time t .)

$$dD/dt = \underline{\hspace{2cm}} \text{ mph.}$$

8.(1 pt)

This is problem 28 Section 3.7 page 163.

One end of a rope is fastened to a boat and the other end is wound around a windlass located on a dock at a point 4 meters above the level of the boat. (see picture in the book) If the boat is drifting away from the dock at the rate of 2 meters/min, how fast is the rope unwinding at the instant when the length of the rope is 5 meters.

(Let $D(t)$ denote the length of rope at time t .)

$$dD/dt = \underline{\hspace{2cm}} \text{ meters/min.}$$

9.(1 pt)

This is problem 32 Section 3.7 page 163.

Consider a piece of ice in the shape of a sphere that is melting at the rate of 5 cubic in./min. Model the volume of ice by a function of the radius r . How fast is the radius changing at the instant when the radius is 4 inches

$$dr/dt = \underline{\hspace{2cm}} \text{ in/min.}$$

How fast is the surface area of the sphere changing at the same instant?

$$dS/dt = \underline{\hspace{2cm}} \text{ sq. in/min.}$$

10.(1 pt)

This is problem 34 Section 3.7 page 163.

Model a water tank by a cone 40 ft high with a circular base of radius 20 feet at the top. Water is flowing into the tank at a constant rate of 80 cubic ft/min. How fast is the water level rising when the water is 12 ft deep?

(Let $x(t)$ denote the radius of the top circle of water at time t and $y(t)$ denote the height of the water at time t .)

$$dy/dt = \underline{\hspace{2cm}} \text{ ft/min.}$$

11.(1 pt)

This is problem 40 Section 3.7 page 164.

A person 6 ft tall is watching a streetlight 18 ft high while walking toward it at a speed of 5 ft/sec (see figure 3.53 in the book). At what rate is the angle of elevation of the person's line of sight changing with respect to time when the person is 9 ft from the base of the light?

(Let $x(t)$ denote the distance from the lamp at time t and θ the angle of elevation at time t)

$$d\theta/dt = \underline{\hspace{2cm}} \text{ radians/sec.}$$

12.(1 pt)

This is problem 44 Section 3.7 page 164.

A swimming pool is 60 ft long and 25 ft wide. Its depth varies uniformly from 3 ft at the shallow end to 15 ft at the deep end (see figure 3.56 in the book). Suppose the pool is being filled with water at the rate of 800 cubic ft/min. At what rate is the depth of the water increasing when it is 5 ft deep at that end?

(Let $y(t)$ denote the depth of the water at the deep end at time t and 25 by x the dimensions of the horizontal rectangular surface at time t)

$$dy/dt = \text{_____} \text{ ft/min} .$$

13.(1 pt)

This is problem 2 Section 3.8 page 173.

Find the differential

$$d(3 - 5x^2) = \text{_____} dx .$$

14.(1 pt)

This is problem 6 Section 3.8 page 173.

Find the differential

$$d(x \sin(2x)) = \text{_____} dx .$$

15.(1 pt)

This is problem 12 Section 3.8 page 173.

Find the differential

$$d\left(\frac{\sin^{-1}(x)}{e}\right) = \text{_____} dx .$$

16.(1 pt)

This is problem 14 Section 3.8 page 173.

Find the differential

$$d\left(x\sqrt{x^2 - 1}\right) = \text{_____} dx .$$

17.(1 pt)

This is problem 19 Section 3.8 page 173.

In this problem we use differentials to approximate $\sqrt{0.99}$.

Thus we use the formula

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x$$

For this problem we have

$$f(x) = \text{_____} .$$

$$f'(x) = \text{_____} .$$

$$x_0 = \text{_____} .$$

$$\Delta x = \text{_____} .$$

$$f(x_0 + \Delta x) = \text{_____} .$$

18.(1 pt)

This is problem 31 Section 3.8 page 174.

A soccer ball made of leather $1/8$ in thick has an inner diameter of 8.5 in . Use the formula $V = (4/3)\pi r^3$ to calculate the volume. Model the ball as a hollow sphere and estimate the volume of its leather shell.

$$dV = \text{_____} \text{ in}^3 .$$

19.(1 pt)

This is problem 32 Section 3.8 page 174.

A cubical box is to be constructed from three kinds of building materials. The material used in the four sides of the box costs 2 cents/sq.in. , the material for the bottom costs 3 cents/sq.in. , and the material for the lid costs 4 cents/sq.in. . Estimate the additional cost of all the building materials if the length of a side is increased from 20 in. to 21 in. .

Let $C(x)$ denote the total cost (in cents) for materials when the side has length x .

$$\Delta C \approx \text{_____} \text{ cents} .$$

Find the actual change in cost

$$C(21) - C(20) = \text{_____} \text{ cents} .$$

20.(1 pt)

This is problem 34 Section 3.8 page 174.

A drug is injected into a patient's bloodstream. The concentration of the drug in the bloodstream t hours after the blood is injected is modeled by the formula

$$C(t) = \frac{0.12t}{t^2 + t + 1}$$

milligrams per cubic centimeter. Estimate the change in concentration over the time period from 30 to 35 minutes after the injection.

Let $C(x)$ denote the total cost (in cents) for materials when the side has length x .

$$\Delta C \approx \text{_____} \text{ mg/cc}^3 .$$

21.(1 pt)

This is problem 42 Section 3.8 page 175.

Suppose the total cost of manufacturing q units of a product is

$$C(q) = 3q^2 + q + 500$$

dollars.

(a) Use marginal analysis to estimate the cost of manufacturing the 41st unit.

(NOTE: if $C(q)$ is the cost to produce q units of a product then the Marginal Cost $MC(q)$ is the additional cost to produce one more unit. See page 170.)

$$MC(40) = \text{_____} \text{ dollars} .$$

(b) Compute the actual cost of manufacturing the 41st unit.

$$C(41) - C(40) = \text{_____} \text{ dollars} .$$

22.(1 pt)

This is problem 44 Section 3.8 page 175.

Suppose the total cost (in dollars) of producing x units of a particular commodity is modeled by

$$C(x) = \frac{1}{7}x^2 + 4x + 100$$

and that each unit of the commodity can be sold for

$$p(x) = \frac{1}{4}(80 - x)$$

dollars.

(a) What is the marginal cost?. (See page 170.) $C'(x) = \text{_____}$ dollars.

(b) What is the price when the marginal cost is 10 dollars? $p = \text{_____}$.

(c) Estimate the cost of producing the 11th unit. $C = \text{_____}$.

(d) Find the actual cost of producing the 11th unit. $C(11) - C(10) = \text{_____}$.